DTU

Model-based Machine Learning

Mixture models

Outline

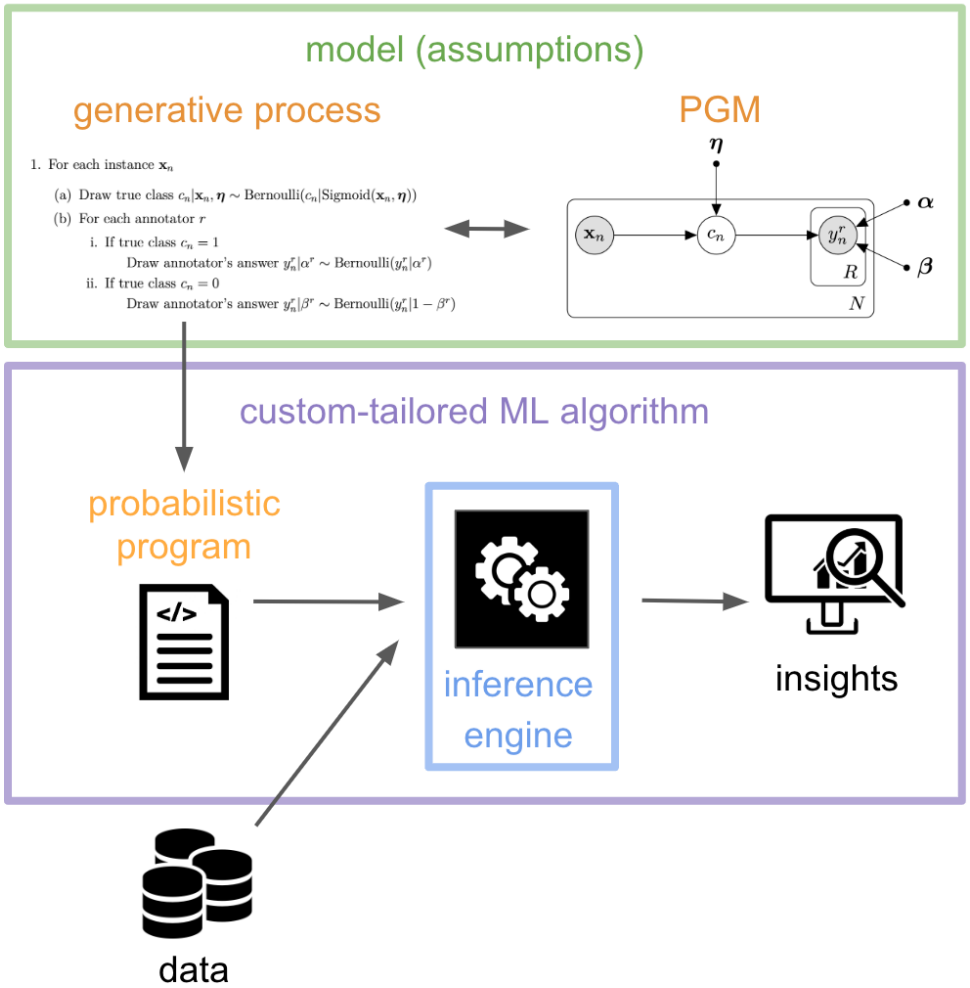
Mixture models

Road to MBML: where are we?

Foundations of PGMs and probabilistic programming
Probabilistic modelling in various contexts - your modelling toolbox
Bayesian inference (exact and approximate)

Week	Topic
1	Intro to the course + Prob. review
2	PGM foundations
3	PGM foundations II
4	Freq. vs Bayesian + Prob. Prog. + Mixture models
5	Regression models
6	Classification and Hierarchical models
7	Temporal models
8	Topic Models
9	Markov-chain Monte Carlo (MCMC)
10	Variational inference
11	Generative models
12	Gaussian processes
13	Project support

Model-based Machine Learning



Learning objectives

At the end of this lecture, you should be able to:

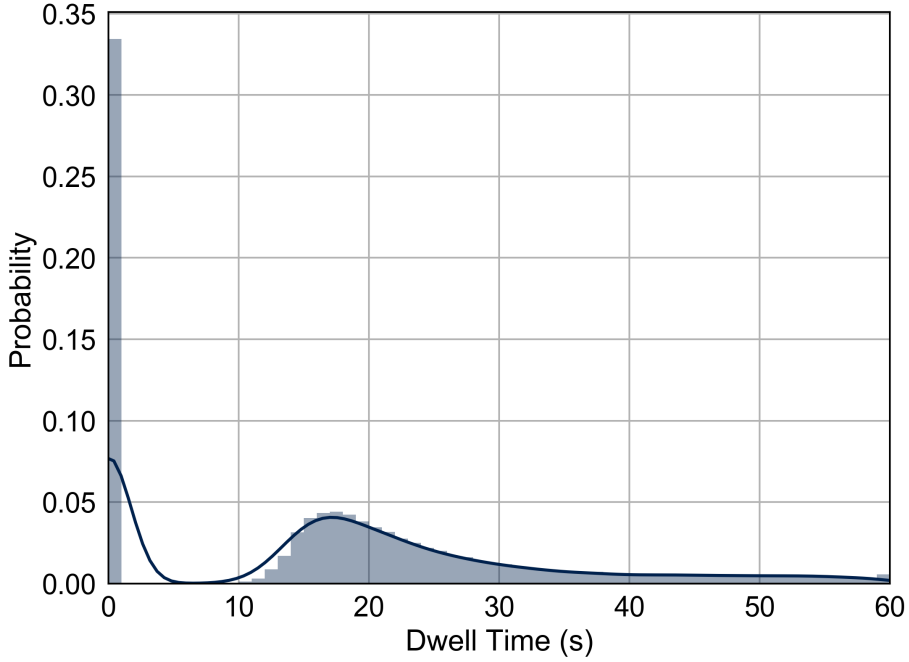
- Understand the concept mixture model, its representation, and inference challenges
- Implement mixture models in Pyro and apply them to model data

Mixture models

- A PGM is composed of observed and latent variables, parameters, constants.
- In this course, we'll approach some examples from this very large family
- Mixture models are pervasive in data modelling in general

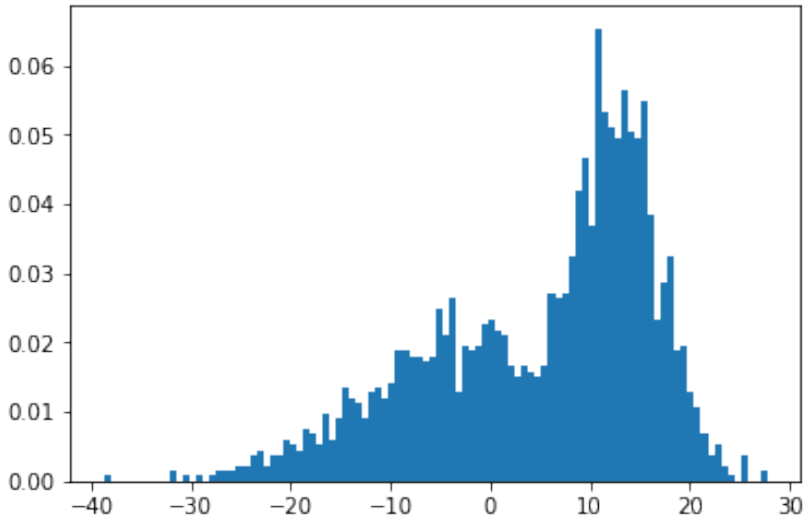
Mixture models

- A PGM is composed of observed and latent variables, parameters, constants.
- In this course, we'll approach some examples from this very large family
- Mixture models are pervasive in data modelling in general
- Problem:
  - Sub-populations of data
  - Data generated from combination/competition of multiple sources
  - Number of sources usually discrete and finite



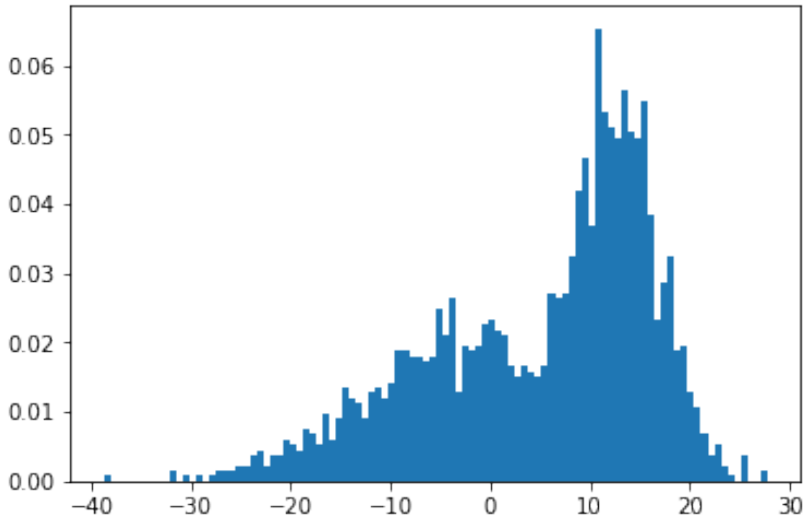
The canonical example: Gaussian Mixture

- What we observe

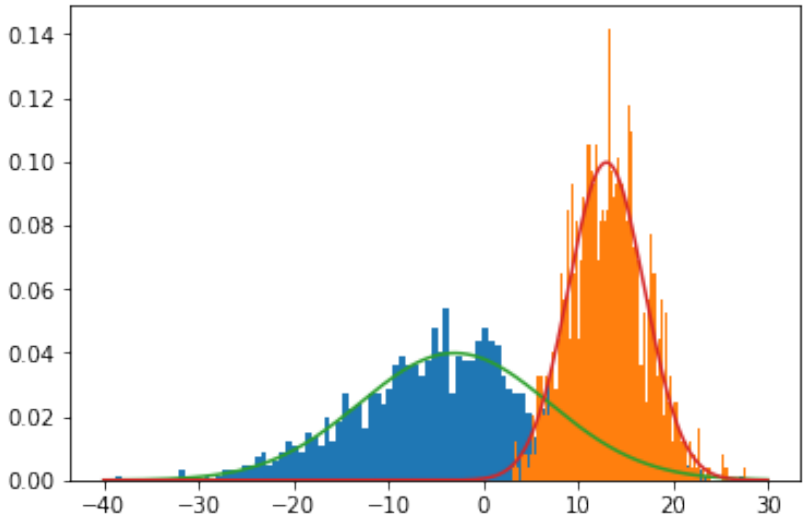


The canonical example: Gaussian Mixture

- What we observe

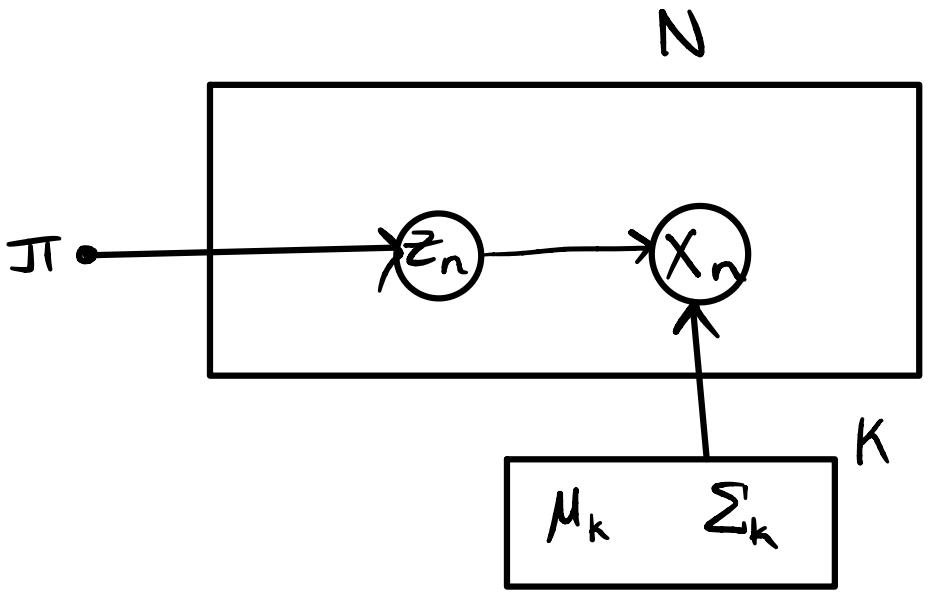


- What really happens



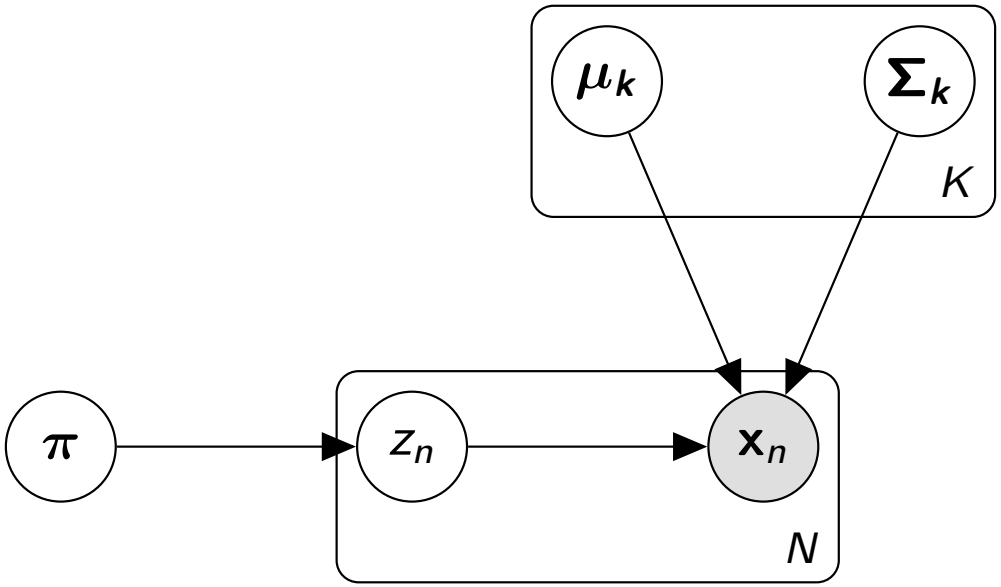
Generative story

1. Draw π, and $(\boldsymbol{\mu_k}, \boldsymbol{\Sigma_k})$ for all K gaussians
2. For $n = 1, 2, ..., N$
   1. Draw $z_n \sim \text{Categorical}(\pi)$
      where π is a vector $(1 \times K)$ with the probabilities of each class
   2. Generate $\mathbf{x}_n$, from the k-th Gaussian, $\mathbf{x}_n \sim \mathcal{N}(\boldsymbol{\mu}_{z_n}, \boldsymbol{\Sigma}_{z_n})$



Generative story

1. Draw π, and $(\boldsymbol{\mu_k}, \boldsymbol{\Sigma_k})$ for all K gaussians π sums to 1
2. For $n = 1, 2, ..., N$
   1. Draw $z_n \sim \text{Categorical}(\pi)$
      where π is a vector $(1 \times K)$ with the probabilities of each class
   2. Generate $\mathbf{x}_n$, from the k-th Gaussian, $\mathbf{x}_n \sim \mathcal{N}(\boldsymbol{\mu}_{z_n}, \boldsymbol{\Sigma}_{z_n})$



Generative story

1. Draw $\boldsymbol{\pi}$
2. Draw $(\boldsymbol{\mu_k}, \boldsymbol{\Sigma_k})$ for all K Gaussians
3. For $n = 1, 2, ..., N$ rolling a dice
   1. Draw $z_n \sim \text{Categorical}(\boldsymbol{\pi})$
      where $\boldsymbol{\pi}$ is a vector $(1 \times K)$ with the probabilities of each class
   2. Generate $\mathbf{x}_n$, from the k-th Gaussian, $\mathbf{x}_n \sim \mathcal{N}(\boldsymbol{\mu}_{z_n}, \boldsymbol{\Sigma}_{z_n})$

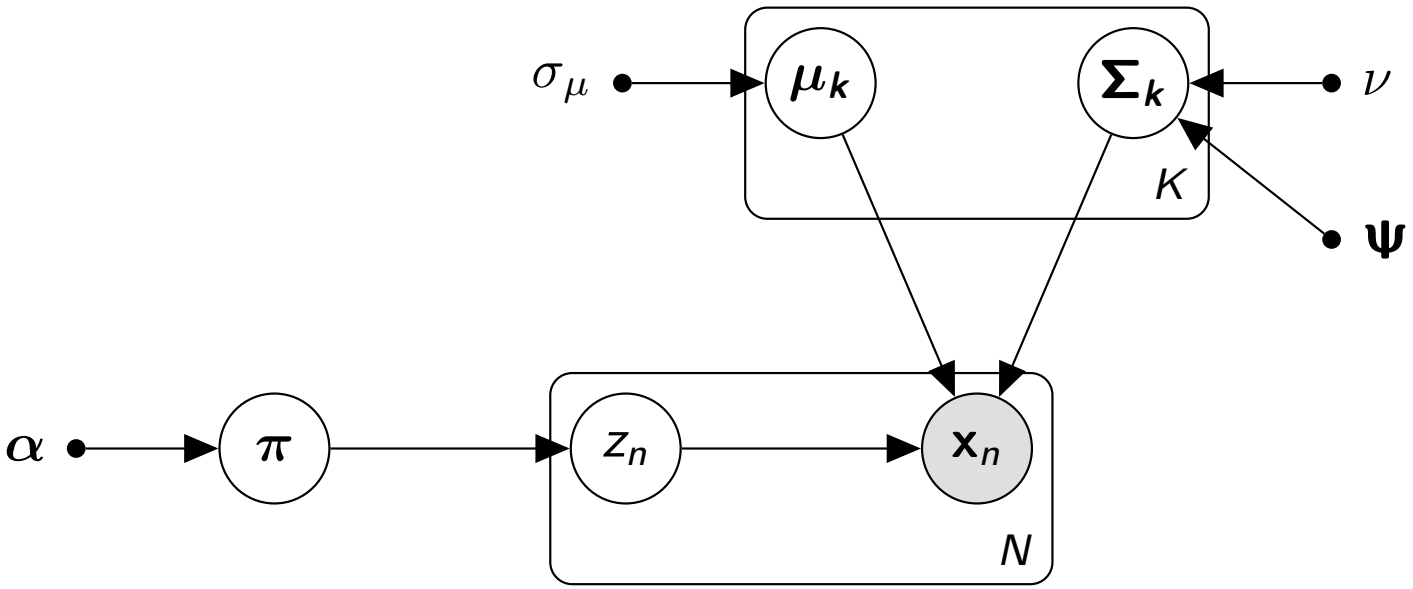
Factorization:

$$p(\boldsymbol{\pi})\left(\prod_{k=1}^{K} p(\boldsymbol{\mu_k})p(\boldsymbol{\Sigma_k})\right)\prod_{n=1}^{N} p(z_n|\boldsymbol{\pi})p(\mathbf{x}_n|\boldsymbol{\mu}_{z_n}, \boldsymbol{\Sigma}_{z_n})$$

Note: in practice we need to be exhaustive

...particularly in probabilistic programming (e.g. Pyro)

- $\pi \sim \text{Dirichlet}(\alpha)$
- $\mu_k \sim \mathcal{N}(\mathbf{0}, \sigma_\mu \mathbf{I})$ Normal gaussian prior $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ $\begin{bmatrix} \sigma_\mu & 0 \\ 0 & \sigma_\mu \end{bmatrix}$
- $\boldsymbol{\Sigma}_k \sim \mathcal{W}^{-1}(\boldsymbol{\Psi}, \nu)$
  $\mathcal{W}^{-1}$ denotes the Inverse Wishart distribution; typically ν = degrees of freedom (typically number of dimensions of $\mathbf{x}$), and $\boldsymbol{\Psi} = \mathbf{I}$



Note: in practice we need to be exhaustive

...particularly in probabilistic programming (e.g. Pyro)

- $\boldsymbol{\pi} \sim \text{Dirichlet}(\boldsymbol{\alpha})$
- $\boldsymbol{\mu_k} \sim \mathcal{N}(\mathbf{0}, \sigma_\mu \mathbf{I})$
- $\boldsymbol{\Sigma_k} \sim \mathcal{W}^{-1}(\boldsymbol{\Psi}, \nu)$
  $\mathcal{W}^{-1}$ denotes the Inverse Wishart distribution; typically $\nu =$ degrees of freedom (typically number of dimensions of $\mathbf{x}$), and $\boldsymbol{\Psi} = \mathbf{I}$

The factorization becomes:

$$\text{Dirichlet}(\boldsymbol{\pi}|\boldsymbol{\alpha})\left(\prod_{k=1}^{K}\mathcal{N}(\boldsymbol{\mu_k}|\mathbf{0}, \sigma_\mu \mathbf{I})\mathcal{W}^{-1}(\boldsymbol{\Sigma_k}|\boldsymbol{\Psi}, \nu)\right)\prod_{n=1}^{N}\pi_{z_n}\mathcal{N}(\mathbf{x}|\boldsymbol{\mu_{z_n}}, \boldsymbol{\Sigma_{z_n}})$$

Full generative process

1. Draw mixture components $\boldsymbol{\pi} \sim \text{Dirichlet}(\boldsymbol{\alpha})$
2. For each mixture component (Gaussian) $k \in \{1, \ldots, K\}$
   1. Draw mean vector $\boldsymbol{\mu_k} \sim \mathcal{N}(\mathbf{0}, \sigma_\mu \mathbf{I})$
   2. Draw covariance matrix $\boldsymbol{\Sigma_k} \sim \mathcal{W}^{-1}(\boldsymbol{\Psi}, \nu)$
3. For $n = 1, 2, ..., N$
   1. Draw $z_n \sim \text{Categorical}(\boldsymbol{\pi})$
      where $\boldsymbol{\pi}$ is a vector $(1 \times K)$ with the probabilities of each class
   2. Generate $\mathbf{x}_n$, from the k-th Gaussian, $\mathbf{x}_n \sim \mathcal{N}(\boldsymbol{\mu}_{z_n}, \boldsymbol{\Sigma}_{z_n})$

Note that the concept of mixture models **can be extended to other distributions beyond Gaussians!**

Playtime!

- Gaussian mixture models - “K-means clustering” (Part 3 of the notebook)
  - See “03 - Probabilistic Programming with Pyro.ipynb" notebook
  - Expected duration: 30 minutes

Readings

- **Main reading**: Chapter 9.2: "Mixture Models and EM", pages 430-435 of Chris Bishop's book, "Pattern Recognition and Machine Learning" (PRML) URL: https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf)
- More on Mixture Models: Chapter 11: "Mixture models and the EM algorithm", pages 337-345 of Kevin Murphy's book "Machine Learning: A Probabilistic Perspective"